
Problem Set 9

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PY 451 Section A1

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1 Wavefunction Spreading

1. Since the wavefunction in momentum space $\psi(p)$ is the fourier transform of the wavefunction $\psi(x)$ in position space (where I mildly abuse notation and use ψ for both functions),

$$\psi(p) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \psi(x) dx = \frac{1}{\sqrt{2\pi\hbar}}.$$

Also recall the decomposition

$$|\psi, t=0\rangle = \mathbb{1}|\psi, t=0\rangle = \int_{-\infty}^{\infty} |p\rangle \langle p| \psi, t=0\rangle dp = \int_{-\infty}^{\infty} \psi(p) |p\rangle dp.$$

Therefore,

$$|\psi, t=0\rangle = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} |p\rangle dp.$$

2. We first find the time evolution for an eigenstate. Assume $|p, t\rangle$ is separable; that is, $|p, t\rangle = \phi(t)|p\rangle$ for some time evolution $\phi(t)$. We then have

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |p, t\rangle &= H|p, t\rangle \implies i\hbar \frac{\partial}{\partial t} f(t)|p\rangle = f(t) \frac{P^2}{2m} |p\rangle \\ &\implies \left(\frac{\partial}{\partial t} f(t) \right) |p\rangle = \frac{f(t)}{i\hbar} \frac{p^2}{2m} |p\rangle \\ &\implies \frac{\partial}{\partial t} f(t) = -\frac{ip^2}{2m\hbar} f(t) \\ &\implies f(t) = \exp\left(-i\frac{p^2}{2m\hbar}t\right) = \exp\left(-\frac{iE_p}{\hbar}t\right), \end{aligned}$$

where we have set $E_p = p^2/2m$. Applying this time evolution to the $|p\rangle$ states in our original equation, we have

$$|\psi, t\rangle = \int_{-\infty}^{\infty} \psi(p, 0) |p, t\rangle dp = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{-iE_p t/\hbar} |p\rangle dp.$$

We can set $\psi(p, t)$ equal to $\psi(p, 0)$ times the time evolution in the previous problem, and thus we obtain

$$\psi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} e^{-iE_p t/\hbar}.$$

3. Fourier transforming back gives gives

$$\begin{aligned}
\psi(x, t) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \psi(p, t) dp \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\hbar} e^{-iE_p t/\hbar} e^{ipx/\hbar} dp \\
&= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar} \left(\frac{p^2}{2m}t - px\right)\right) dp \\
&= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar} \left(\frac{t}{2m}p^2 - xp + \frac{x^2m}{2t} - \frac{x^2m}{2t}\right)\right) dp \\
&= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar} \left(\left(\sqrt{\frac{t}{2m}}p - \frac{x}{2}\sqrt{\frac{2m}{t}}\right)^2 - \frac{x^2m}{2t}\right)\right) dp \\
&= \frac{1}{2\pi\hbar} \exp\left(\frac{ix^2m}{2t\hbar}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar} \left(\sqrt{\frac{t}{2m}}p - \frac{x}{2}\sqrt{\frac{2m}{t}}\right)^2\right) dp
\end{aligned}$$

$$\begin{aligned}
\text{Let } u &= \sqrt{\frac{t}{2m}}p - \frac{x}{2}\sqrt{\frac{2m}{t}} \\
&= \frac{1}{2\pi\hbar} \exp\left(\frac{ix^2m}{2t\hbar}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{i}{\hbar}u^2\right) \sqrt{\frac{2m}{t}} du \\
&= \frac{1}{2\pi\hbar} \sqrt{\frac{2m}{t}} \exp\left(\frac{ix^2m}{2t\hbar}\right) \int_{-\infty}^{\infty} e^{-iu^2/\hbar} du
\end{aligned}$$

$$\begin{aligned}
\text{Let } v &= \sqrt{\frac{i}{\hbar}}u \\
&= \frac{1}{2\pi\hbar} \sqrt{\frac{2m}{t}} \exp\left(\frac{ix^2m}{2t\hbar}\right) \int_{-\infty}^{\infty} \sqrt{\frac{\hbar}{i}} e^{-v^2} dv \\
&= \frac{1}{2\pi\hbar} \sqrt{\frac{2m}{t}} \exp\left(\frac{ix^2m}{2t\hbar}\right) \sqrt{-i\hbar}\sqrt{\pi} \\
&= \frac{1}{2\pi\hbar} \sqrt{-\frac{i2\pi m\hbar}{t}} \exp\left(\frac{ix^2m}{2t\hbar}\right) \\
&= \sqrt{-\frac{i2\pi m\hbar}{4\pi^2\hbar^2t}} \exp\left(\frac{ix^2m}{2t\hbar}\right) \\
&= \sqrt{-\frac{im}{2\pi\hbar t}} \exp\left(\frac{ix^2m}{2t\hbar}\right),
\end{aligned}$$

where I have used the well-known value of the Gaussian integral to complete the problem. There is an important step that must be explained, however. When I make the substitution $v = \sqrt{\frac{i}{\hbar}}u$, the integral becomes a contour integral along the straight line C in the complex plane passing through the origin at an angle of $\pi/2$. However, since the integrand is holomorphic everywhere and thus has no singularities, and since the path C is clearly homotopic to $\mathbb{R}$ when raised to the surface defined by the integrand (i.e. a continuous deformation $C \rightarrow \mathbb{R}$ exists without passing singularities, since there are none), it follows that integrating over $\mathbb{R}$ rather than C will preserve the value of the integral. Hence, keeping the bounds at $\pm\infty$ is justified.

4. First, we compute $\langle X \rangle$.

$$\begin{aligned}
\langle \psi, t | X | \psi, t \rangle &= \int_{-\infty}^{\infty} x \langle \psi, t | x \rangle \langle x | \psi, t \rangle dx \\
&= \int_{-\infty}^{\infty} x |\psi(x, t)|^2 dx \\
&= \int_{-\infty}^{\infty} x \left| \sqrt{-i} \right|^2 \left| \sqrt{\frac{m}{2\pi\hbar t}} \right|^2 \left| \exp\left(\frac{ix^2 m}{2t\hbar}\right) \right|^2 dx \\
&= \int_{-\infty}^{\infty} x \left| i \cdot e^{i\pi/4} \right|^2 \frac{m}{2\pi\hbar t} dx.
\end{aligned}$$

This is an odd integral, so it equals 0. Now we compute $\langle X^2 \rangle$.

$$\begin{aligned}
\langle \psi, t | X^2 | \psi, t \rangle &= \langle \psi, t | \int_{-\infty}^{\infty} x X | x \rangle \langle x | dx | \psi, t \rangle \\
&= \int_{-\infty}^{\infty} x^2 \langle \psi, t | x \rangle \langle x | \psi, t \rangle dx \\
&= \int_{-\infty}^{\infty} x^2 |\psi(x, t)|^2 dx \\
&= \frac{x^3 m}{6\pi\hbar t} \Big|_{-\infty}^{\infty}.
\end{aligned}$$

If $t \neq 0$, then clearly $\langle X^2 \rangle \rightarrow \infty$. Therefore, $(\Delta x)^2 \rightarrow \infty$ for all $t > 0$, and consequently the same for Δx . When $t = 0$, the particle's location is precisely defined to be 0, so there cannot be any deviation. Thus,

$$\Delta x = \begin{cases} 0, & t = 0 \\ \infty, & t > 0 \end{cases}.$$

2 Finite Potential Well and Scattering

1. For all particles to be transmitted, from our notes we must have $\xi = 0$, and thus $k_2 = n\pi/a$ for some $n \in \mathbb{Z}$. Recall that

$$k_2 = \sqrt{\frac{2m}{\hbar^2}(E + V_0)}.$$

Setting $E = 11\text{eV}$ and $V_0 = 8\text{eV}$, we find

$$\frac{n\pi}{a} = \sqrt{\frac{38m}{\hbar^2}} \implies a = \frac{n\pi\hbar}{\sqrt{38m}}.$$

We now need to find the proper n , using the condition on the number of bound states. Recalling our solutions to finite potential wells, recall that bound states with energy E must satisfy

$$k_1 = k_2 \tan k_2 a, \quad \text{or} \quad k_1 = -k_2 \cot k_2 a.$$

More explicitly,

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}}, \quad k_2 = \sqrt{\frac{2m(E + 8)}{\hbar^2}}.$$

We end up with

$$\sqrt{\frac{2mE}{\hbar^2}} = \sqrt{\frac{2m(E+8)}{\hbar^2}} \tan\left(\frac{n\pi\hbar}{\sqrt{38m}} \sqrt{\frac{2m(E+8)}{\hbar^2}}\right),$$

$$\sqrt{\frac{2mE}{\hbar^2}} = -\sqrt{\frac{2m(E+8)}{\hbar^2}} \cot\left(\frac{n\pi\hbar}{\sqrt{38m}} \sqrt{\frac{2m(E+8)}{\hbar^2}}\right).$$

I plugged these into Mathematica with the following code. It iterates n over $\{1, \dots, 12\}$, and plots k_1 minus both the even and odd parity functions.

```

1 hbar = 1.054571817*10^-34;      (* Js *)
2 m    = 9.10938356*10^-31;      (* kg *)
3 eV   = 1.602176634*10^-19;     (* J/eV *)
4 V0   = 8*eV;                   (* well depth (J) *)
5 Eres = 11*eV;                  (* resonance energy (J) *)
6
7 k1[x_] := Sqrt[2 m x]/hbar
8 k2[x_] := Sqrt[2 m (x+V0)]/hbar
9 a[n_] := (n*Pi*hbar)/(Sqrt[2*(V0+Eres)*m])
10
11 fEven[x_, n_] := k1[x] - (k2[x]*Tan[a[n]*k2[x]])
12 fOdd[x_, n_] := k1[x] + (k2[x]*Cot[a[n]*k2[x]])
13
14 Do[
15   Print[
16     Plot[
17       {fEven[x, n], fOdd[x, n], 0},
18       {x, 0, V0},
19       Exclusions -> {Tan[_], Cot[_]},
20       ExclusionsStyle -> None,
21       PlotRange -> {{0, V0}, Automatic},
22       PlotStyle -> {Blue, Red, Gray},
23       PlotLegends -> {"even (tan)", "odd (cot)", "zero"},
24       PlotLabel -> Row[{"n = ", n}],
25       AxesLabel -> {"e (eV)", "f(e)"}
26     ]
27   ],
28   {n, 1, 12}
29 ]
30
31 Manual inspection of the roots of the plots shows that $n$ in \left\{
9,10,11 \right\} $ yield 5 bound states.

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This uses the mass of an electron for this calculation. Unit independence should make this immaterial. I also set $\hbar = m = 1$ and found the same thing, but my units for V_0 felt off, so I redid it. Manual inspection of the plots shows there are 5 roots for $n \in \{9, 10, 11\}$, so we can take $n = 9$ to be the smallest value. An image of the output is included on the next page. Plugging in, we have

$$L = 2a = \frac{9\pi\hbar}{\sqrt{19m}} \approx 2.53 \cdot 10^{-9}.$$

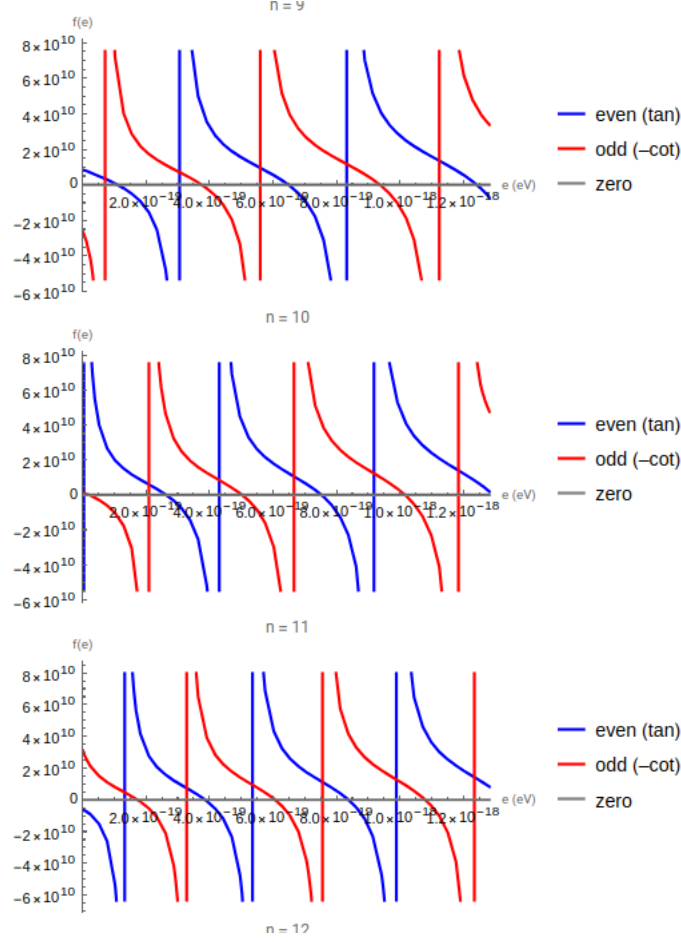


Figure 1: Mathematica Output

2. This occurs when $k_2 = n\pi/a$. More precisely,

$$\sqrt{\frac{2m(E_n + V_0)}{\hbar^2}} = \frac{n\pi}{\frac{9\pi\hbar}{\sqrt{38m}}} = \frac{n\sqrt{38m}}{9\hbar}.$$

Simplifying and setting $V_0 = 8$, we have

$$\sqrt{E_n + 8} = \frac{n\sqrt{19}}{9}.$$

Solving for E_n gives us

$$E_n = \frac{19n^2}{81} - 8.$$

All E_n s will be transmitted with unit probability, since they imply $\xi = 0$ and thus $T = 1$. Note that choosing $n = 9$ recovers $E_9 = 11$.